

```

> restart;
> # Ex1:
> # because of round errors,
> evalf(25^(1/8));
1.495348781

```

(1)

```

> (%)^8;
24.99999997

```

(2)

```

> # you cannot represent in finite memory an infinite nr of
> # decimals

```

```

> # Ex2: fixed point vs equilibrium point (seminar 5)

```

```

> # f(fixed_point) = fixed_point -> fixed_point is a number

```

```

> # a) fixed points of f1, f1(x)=x*(1-x)

```

```

> solve(x*(1-x)=x,x);
0, 0

```

(3)

```

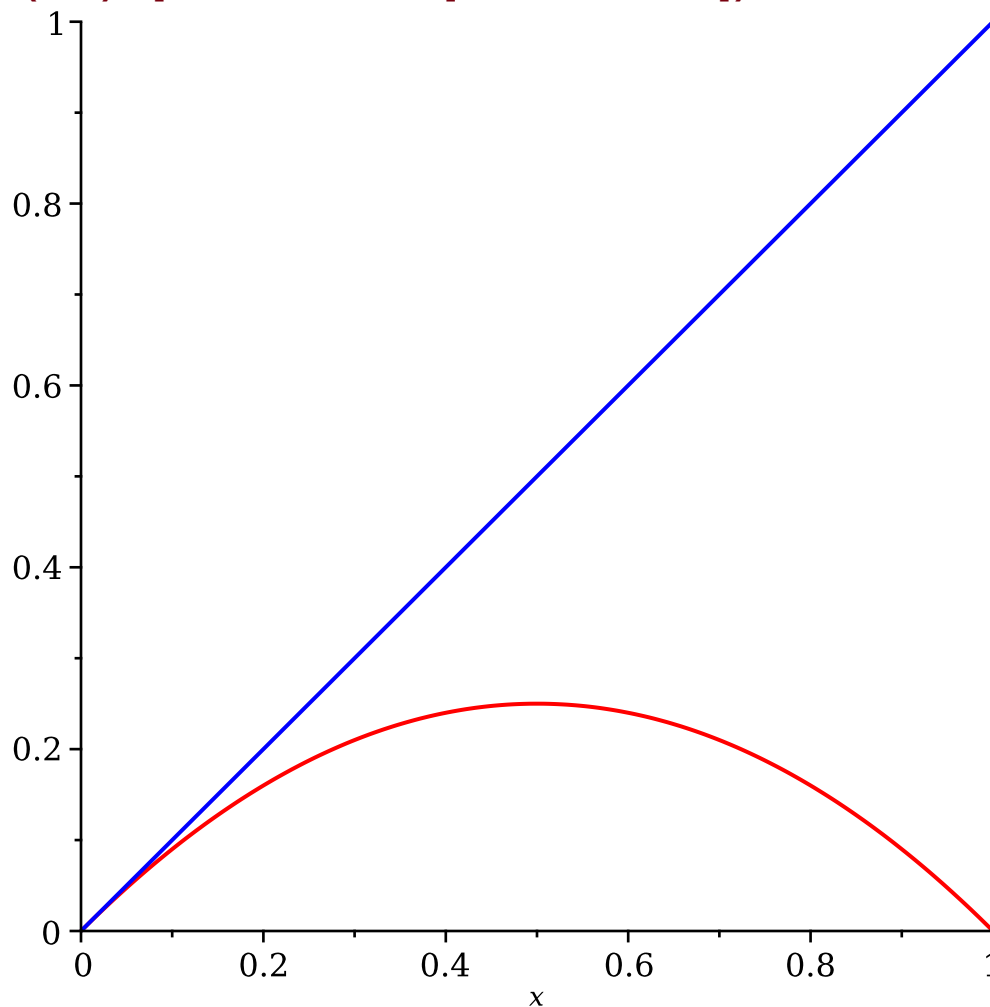
> # b)

```

```

> plot([x*(1-x),x],x=0..1,color=["Red","Blue"]);

```



```

> # there are no other fixed points apart from 0 because the graphs
> # don't intersect in another point

```

```

> # c) compute the first 200 iterations of f1
> restart;
> with(plots)

```

(4)

```

[animate, animate3d, animatecurve, arrow, changecoords, complexplot,
complexplot3d, conformal, conformal3d, contourplot, contourplot3d,
coordplot, coordplot3d, densityplot, display, dualaxisplot, fieldplot,
fieldplot3d, gradplot, gradplot3d, implicitplot, implicitplot3d, inequal,
interactive, interactiveparams, intersectplot, listcontplot, listcontplot3d,
listdensityplot, listplot, listplot3d, loglogplot, logplot, matrixplot, multiple,
odeplot, pareto, plotcompare, pointplot, pointplot3d, polarplot,
polygonplot, polygonplot3d, polyhedra_supported, polyhedraplot,
rootlocus, semilogplot, setcolors, setoptions, setoptions3d, shadebetween,
spacecurve, sparsematrixplot, surfdata, textplot, textplot3d, tubeplot]

```

```

> x:=0.5

```

x := 0.5

(5)

```

> for i from 1 to 200 do x:=x*(1-x); f[i]:=x: od:
> # od is the closing of do

```

```

> points:=[[n,f[n]]$n=1..200];
points := [[1, 0.25], [2, 0.1875], [3, 0.15234375], [4, 0.1291351318], [5,
0.1124592495], [6, 0.09981216670], [7, 0.08984969808], [8,
0.08177672983], [9, 0.07508929629], [10, 0.06945089387], [11,
0.06462746721], [12, 0.06045075769], [13, 0.05679646358], [14,
0.05357062530], [15, 0.05070081340], [16, 0.04813024092], [17,
0.04581372083], [18, 0.04371482382], [19, 0.04180383800], [20,
0.04005627713], [21, 0.03845177179], [22, 0.03697323304], [23,
0.03560621308], [24, 0.03433841067], [25, 0.03315928422], [26,
0.03205974609], [27, 0.03103191877], [28, 0.03006893879], [29,
0.02916479771], [30, 0.02831421228], [31, 0.02751251766], [32,
0.02675557903], [33, 0.02603971802], [34, 0.02536165111], [35,
0.02471843776], [36, 0.02410743659], [37, 0.02352626809], [38,
0.02297278280], [39, 0.02244503405], [40, 0.02194125450], [41,
0.02145983585], [42, 0.02099931130], [43, 0.02055834022], [44,
0.02013569487], [45, 0.01973024866], [46, 0.01934096595], [47,
0.01896689299], [48, 0.01860714996], [49, 0.01826092393], [50,
0.01792746259], [51, 0.01760606867], [52, 0.01729609502], [53,
0.01699694012], [54, 0.01670804415], [55, 0.01642888541], [56,
0.01615897713], [57, 0.01589786459], [58, 0.01564512249], [59,
0.01540035263], [60, 0.01516318177], [61, 0.01493325969], [62,
0.01471025744], [63, 0.01449386577], [64, 0.01428379362], [65,

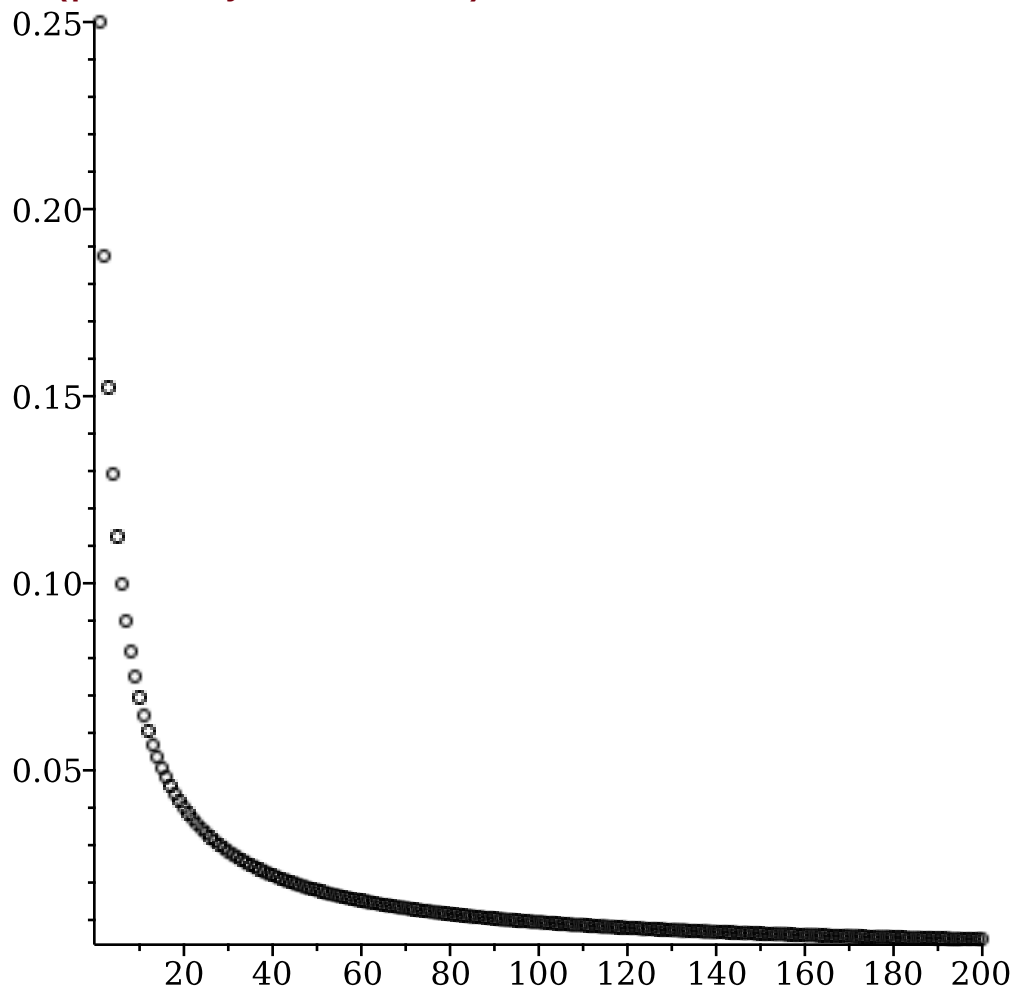
```

(6)

0.01407976686], [66, 0.01388152702], [67, 0.01368883023], [68, 0.01350144616], [69, 0.01331915711], [70, 0.01314175716], [71, 0.01296905138], [72, 0.01280085509], [73, 0.01263699320], [74, 0.01247729960], [75, 0.01232161659], [76, 0.01216979435], [77, 0.01202169045], [78, 0.01187716941], [79, 0.01173610226], [80, 0.01159836616], [81, 0.01146384406], [82, 0.01133242434], [83, 0.01120400050], [84, 0.01107847087], [85, 0.01095573835], [86, 0.01083571015], [87, 0.01071829754], [88, 0.01060341564], [89, 0.01049098322], [90, 0.01038092249], [91, 0.01027315894], [92, 0.01016762115], [93, 0.01006424063], [94, 0.009962951691], [95, 0.009863691285], [96, 0.009766398879], [97, 0.009671016332], [98, 0.009577487775], [99, 0.009485759503], [100, 0.009395779870], [101, 0.009307499190], [102, 0.009220869649], [103, 0.009135845212], [104, 0.009052381544], [105, 0.008970435933], [106, 0.008889967212], [107, 0.008810935695], [108, 0.008733303107], [109, 0.008657032524], [110, 0.008582088312], [111, 0.008508436072], [112, 0.008436042587], [113, 0.008364875772], [114, 0.008294904625], [115, 0.008226099182], [116, 0.008158430474], [117, 0.008091870486], [118, 0.008026392118], [119, 0.007961969148], [120, 0.007898576196], [121, 0.007836188690], [122, 0.007774782837], [123, 0.007714335589], [124, 0.007654824615], [125, 0.007596228275], [126, 0.007538525591], [127, 0.007481696223], [128, 0.007425720445], [129, 0.007370579121], [130, 0.007316253685], [131, 0.007262726117], [132, 0.007209978926], [133, 0.007157995130], [134, 0.007106758236], [135, 0.007056252224], [136, 0.007006461529], [137, 0.006957371026], [138, 0.006908966015], [139, 0.006861232204], [140, 0.006814155697], [141, 0.006767722979], [142, 0.006721920905], [143, 0.006676736684], [144, 0.006632157871], [145, 0.006588172353], [146, 0.006544768338], [147, 0.006501934346], [148, 0.006459659196], [149, 0.006417931999], [150, 0.006376742148], [151, 0.006336079308], [152, 0.006295933407], [153, 0.006256294630], [154, 0.006217153408], [155, 0.006178500412], [156, 0.006140326545], [157, 0.006102622935], [158, 0.006065380929], [159, 0.006028592083], [160, 0.005992248160], [161, 0.005956341122], [162, 0.005920863123], [163, 0.005885806503], [164, 0.005851163785], [165, 0.005816927667], [166, 0.005783091019], [167, 0.005749646877], [168, 0.005716588438], [169, 0.005683909055], [170, 0.005651602233], [171, 0.005619661625], [172, 0.005588081028], [173, 0.005556854379], [174, 0.005525975748], [175, 0.005495439340], [176, 0.005465239487], [177, 0.005435370644], [178, 0.005405827390], [179,

```
0.005376604420], [180, 0.005347696545], [181, 0.005319098687], [182,
0.005290805876], [183, 0.005262813249], [184, 0.005235116046], [185,
0.005207709606], [186, 0.005180589367], [187, 0.005153750861], [188,
0.005127189713], [189, 0.005100901639], [190, 0.005074882442], [191,
0.005049128010], [192, 0.005023634316], [193, 0.004998397414], [194,
0.004973413437], [195, 0.004948678596], [196, 0.004924189176], [197,
0.004899941537], [198, 0.004875932110], [199, 0.004852157396], [200,
0.004828613965]]
```

```
> pointplot(points,symbol=circle)
```



```
> # graph is monotonically decreasing to the fixed point which is 0
```

```
> restart;
```

```
> # Problem 3
```

```
> restart;
```

```
> with(plots);
```

```
[animate, animate3d, animatecurve, arrow, changecoords, complexplot,
complexplot3d, conformal, conformal3d, contourplot, contourplot3d,
coordplot, coordplot3d, densityplot, display, dualaxisplot, fieldplot,
```

(7)

fieldplot3d, gradplot, gradplot3d, implicitplot, implicitplot3d, inequal, interactive, interactiveparams, intersectplot, listcontplot, listcontplot3d, listdensityplot, listplot, listplot3d, loglogplot, logplot, matrixplot, multiple, odeplot, pareto, plotcompare, pointplot, pointplot3d, polarplot, polygonplot, polygonplot3d, polyhedra_supported, polyhedraplot, rootlocus, semilogplot, setcolors, setoptions, setoptions3d, shadebetween, spacecurve, sparsematrixplot, surfdata, textplot, textplot3d, tubeplot]

> x:=0.7

$x := 0.7$

(8)

> restart;

> # Problem 4 and 5

> with(plots);

[animate, animate3d, animatecurve, arrow, changecoords, complexplot, complexplot3d, conformal, conformal3d, contourplot, contourplot3d, coordplot, coordplot3d, densityplot, display, dualaxisplot, fieldplot, fieldplot3d, gradplot, gradplot3d, implicitplot, implicitplot3d, inequal, interactive, interactiveparams, intersectplot, listcontplot, listcontplot3d, listdensityplot, listplot, listplot3d, loglogplot, logplot, matrixplot, multiple, odeplot, pareto, plotcompare, pointplot, pointplot3d, polarplot, polygonplot, polygonplot3d, polyhedra_supported, polyhedraplot, rootlocus, semilogplot, setcolors, setoptions, setoptions3d, shadebetween, spacecurve, sparsematrixplot, surfdata, textplot, textplot3d, tubeplot]

(9)

> lambda:=3.8

$\lambda := 3.8$

(10)

> solve(lambda*x*(1-x)=x,x);

0., 0.7368421053

(11)

> x:=0.7

$x := 0.7$

(12)

> for i from 1 to 200 do: x:=lambda*x*(1-x); f[i]:=x: od:

> points:=[seq(n[n,f[n]],n=1..200)]

points := [1_{1,0.798}, 2_{2,0.6125448}, 3_{3,0.9018679384}, 4_{4,0.3363082083}, 5_{5,0.8481789902}, 6_{6,0.4893312850}, 7_{7,0.9495674784}, 8_{8,0.1819785130}, 9_{9,0.5656768685}, 10_{10,0.9336088859}, 11_{11,0.2355366694}, 12_{12,0.6842247577}, 13_{13,0.8210327068}, 14_{14,0.5583644045}, 15_{15,0.9370556658}, 16_{16,0.2241329110}, 17_{17,0.6608099270}, 18_{18,0.8517326362}, 19_{19,0.4798797801}, 20_{20,0.9484616715}, 21_{21,0.1857520910}, 22_{22,0.5747433564}, 23_{23,0.9287710364}, 24_{24,0.2513905137}, 25_{25,0.7151346287}, 26_{26,0.7741249478},

(13)

27_{27, 0.6644509495}, 28_{28, 0.8472323638}, 29_{29, 0.4918328049}, 30_{30, 0.9497465285},
31_{31, 0.1813666284}, 32_{32, 0.5641965431}, 33_{33, 0.9343394547}, 34_{34, 0.2331271048},
35_{35, 0.6793596596}, 36_{36, 0.8277544274}, 37_{37, 0.5417927342}, 38_{38, 0.9433627960},
39_{39, 0.2030318383}, 40_{40, 0.6148776615}, 41_{41, 0.8998518671}, 42_{42, 0.3424502406},
43_{43, 0.8556766784}, 44_{44, 0.4692775817}, 45_{45, 0.9464133052}, 46_{46, 0.1927176116},
47_{47, 0.5911946284}, 48_{48, 0.9183974511}, 49_{49, 0.2847855771}, 50_{50, 0.7739944583},
51_{51, 0.6647227400}, 52_{52, 0.8468923919}, 53_{53, 0.4927295400}, 54_{54, 0.9497991336},
55_{55, 0.1811868098}, 56_{56, 0.5637609690}, 57_{57, 0.9345512475}, 58_{58, 0.2324278105},
59_{59, 0.6779394689}, 60_{60, 0.8296826726}, 61_{61, 0.5369754745}, 62_{62, 0.9448046942},
63_{63, 0.1981653793}, 64_{64, 0.6038042746}, 65_{65, 0.9090537556}, 66_{66, 0.3141650951},
67_{67, 0.8187684746}, 68_{68, 0.5638693064}, 69_{69, 0.9344987043}, 70_{70, 0.2326013286},
71_{71, 0.6782922120}, 72_{72, 0.8292051713}, 73_{73, 0.5381710297}, 74_{74, 0.9444632955},
75_{75, 0.1993190400}, 76_{76, 0.6064456491}, 77_{77, 0.9069434306}, 78_{78, 0.3207087683},
79_{79, 0.8278476864}, 80_{80, 0.5415603991}, 81_{81, 0.9434363865}, 82_{82, 0.2027838503},
83_{83, 0.6143177293}, 84_{84, 0.9003395356}, 85_{85, 0.3409673737}, 86_{86, 0.8538927703},
87_{87, 0.4740876471}, 88_{88, 0.9474484899}, 89_{89, 0.1892014258}, 90_{90, 0.5829361358},
91_{91, 0.9238620700}, 92_{92, 0.2672955933}, 93_{93, 0.7442249049}, 94_{94, 0.7233459442},
95_{95, 0.7604430390}, 96_{96, 0.6922438090}, 97_{97, 0.8095608080}, 98_{98, 0.5858540033},
99_{99, 0.9219905426}, 100_{100, 0.2733111314}, 101_{101, 0.7547261958}, 102_{102, 0.7034353476},
103_{103, 0.7927334256}, 104_{104, 0.6243671378}, 105_{105, 0.8912246973}, 106_{106, 0.3683842977},
107_{107, 0.8841737661}, 108_{108, 0.3891599663}, 109_{109, 0.9033150504}, 110_{110, 0.3318804865},
111_{111, 0.8425961511}, 112_{112, 0.5039859335}, 113_{113, 0.9499396267}, 114_{114, 0.1807064628},
115_{115, 0.5625962210}, 116_{116, 0.9351105099}, 117_{117, 0.2305796079}, 118_{118, 0.6741680788},
119_{119, 0.8347288251}, 120_{120, 0.5242351318}, 121_{121, 0.9477681019}, 122_{122, 0.1881141623},
123_{123, 0.5803634521}, 124_{124, 0.9254585192}, 125_{125, 0.2621431841}, 126_{126, 0.7350117135},
127_{127, 0.7401240791}, 128_{128, 0.7308936213}, 129_{129, 0.7474149155}, 130_{130, 0.7173862665},
131_{131, 0.7704242024}, 132_{132, 0.6721088528}, 133_{133, 0.8374384627}, 134_{134, 0.5173140787},
135_{135, 0.9488608462}, 136_{136, 0.1843909749}, 137_{137, 0.5714855844}, 138_{138, 0.9305812828},
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143_{143, 0.8899261730}, 144_{144, 0.3722388025}, 145_{145, 0.8879728907}, 146_{146, 0.3780127371},
147_{147, 0.8934526092}, 148_{148, 0.3617411684}, 149_{149, 0.8773610829}, 150_{150, 0.4088747298},
151_{151, 0.9184455034}, 152_{152, 0.2846327706}, 153_{153, 0.7737444345}, 154_{154, 0.6652431414},
155_{155, 0.8462398759}, 156_{156, 0.4944482036}, 157_{157, 0.9498828749}, 158_{158, 0.1809005157},
159_{159, 0.5630669727}, 160_{160, 0.9348857163}, 161_{161, 0.2313227723}, 162_{162, 0.6756876798},

```

163163, 0.8327085888, 164164, 0.5293589807, 165165, 0.9467245911, 166166, 0.1916611309,
167167, 0.5887231388, 168168, 0.9200871775, 169169, 0.2794017005, 170170, 0.7650782831,
171171, 0.6829873146, 172172, 0.8227594421, 173173, 0.5541401016, 174174, 0.9388616277,
175175, 0.2181217926, 176176, 0.6480697695, 177177, 0.8666863047, 178178, 0.4390563850,
179179, 0.9358863280, 180180, 0.2280118144, 181181, 0.6688852222, 182182, 0.8416155704,
183183, 0.5065354479, 184184, 0.9498376941, 185185, 0.1810549861, 186186, 0.5634414968,
187187, 0.9347056707, 188188, 0.2319177235, 189189, 0.6769011935, 190190, 0.8310826773,
191191, 0.5334601911, 192192, 0.9457455792, 193193, 0.1949813388, 194194, 0.5964617420,
195195, 0.9146415030, 196196, 0.2966752111, 197197, 0.7929043147, 198198, 0.6239868373,
199199, 0.8915836040, 200200, 0.3673166681]

```

```
> with(plots)
```

```

[animate, animate3d, animatecurve, arrow, changecoords, complexplot,
complexplot3d, conformal, conformal3d, contourplot, contourplot3d,
coordplot, coordplot3d, densityplot, display, dualaxisplot, fieldplot,
fieldplot3d, gradplot, gradplot3d, implicitplot, implicitplot3d, inequal,
interactive, interactiveparams, intersectplot, listcontplot, listcontplot3d,
listdensityplot, listplot, listplot3d, loglogplot, logplot, matrixplot,
multiple, odeplot, pareto, plotcompare, pointplot, pointplot3d, polarplot,
polygonplot, polygonplot3d, polyhedra_supported, polyhedraplot,
rootlocus, semilogplot, setcolors, setoptions, setoptions3d,
shadebetween, spacecurve, sparsematrixplot, surfdata, textplot,
textplot3d, tubeplot]

```

(14)

```
> pointplot(points);
```

```

Error. (in plots:-pointplot) cannot convert data to an Array of
datatype float[8]

```

```
> restart;
```

```
> #Problem 4 and 5
```

```
> restart:
```

```
> with(plots):
```

```
> lambda:=4;
```

$\lambda := 4$

(15)

```
> solve(lambda*x*(1-x)=x,x); evalf(2/3);
```

0, $\frac{3}{4}$

0.6666666667

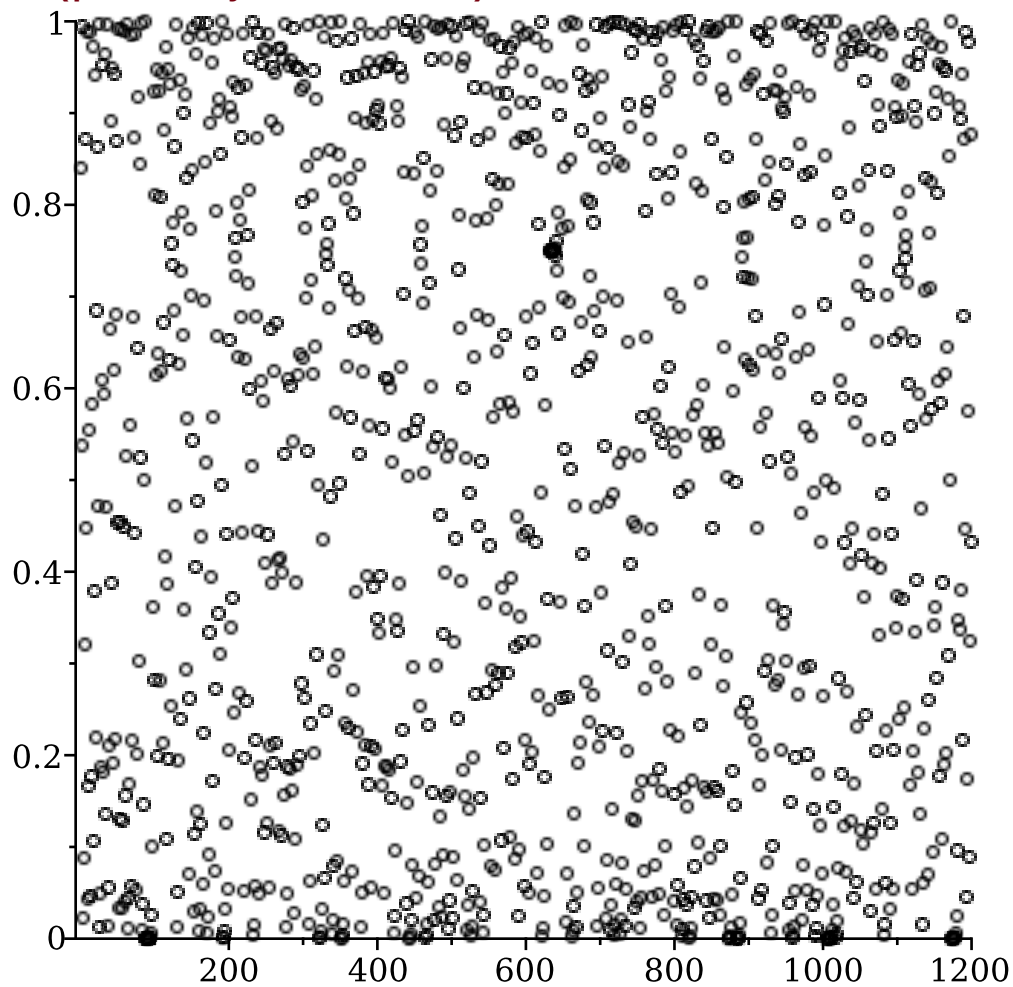
(16)

```
> x:=0.7
```

$x := 0.7$

(17)

```
> for i from 1 to 1200 do x:=lambda*x*(1-x); f[i]:=x: od:  
> points:=[[n,f[n]]$n=1..1200]:  
> pointplot(points,symbol=circle);
```

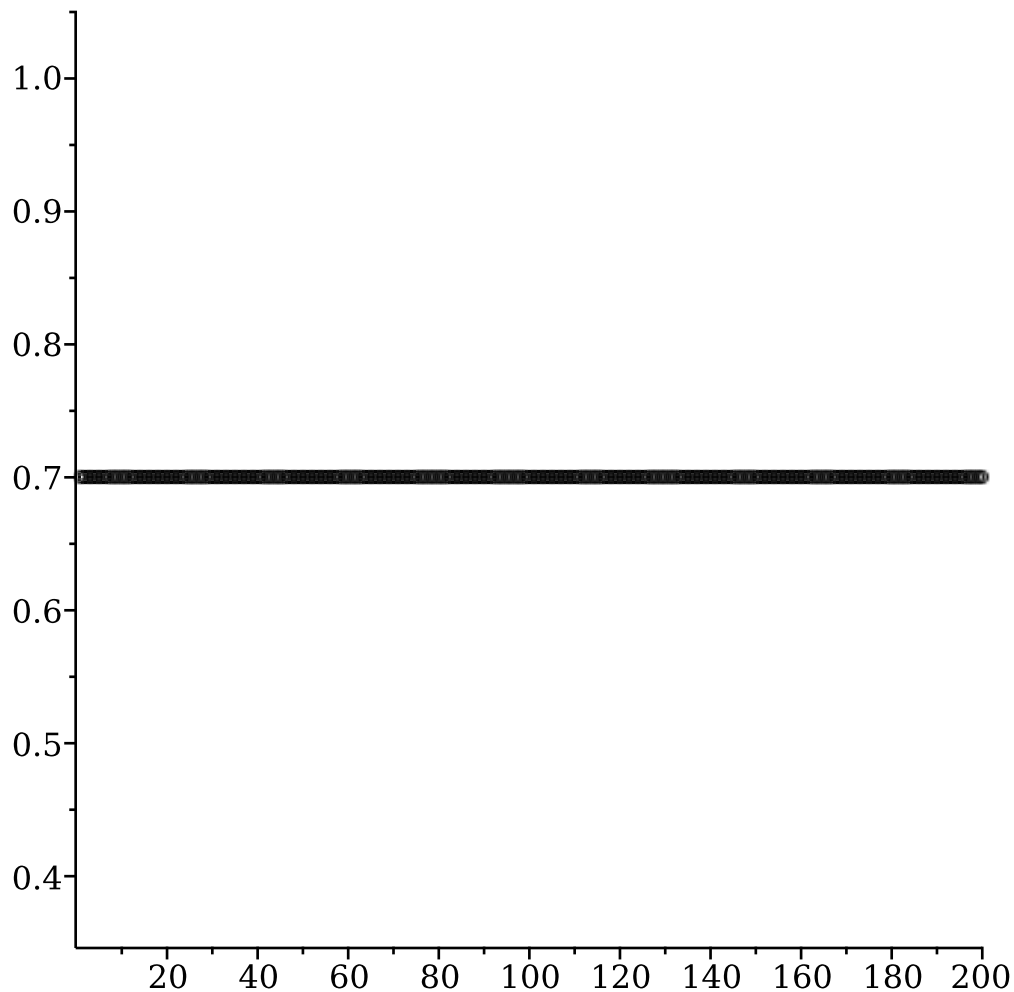


```
> restart;  
> #Problem 3  
> restart;  
> with(plots):  
> x:=0.7
```

$x := 0.7$

(18)

```
> for i from 1 to 200 do x=2*x*(1-x); f[i]:=x: od:  
> points:=[[n,f[n]]$n=1..200]:  
> pointplot(points,symbol=circle);
```



```

> restart;
> #Problem 5 for other lambda values
> lambda:=3.1;
                                 $\lambda := 3.1$ 

```

(19)

```

> solve(lambda*x*(1-x)=x,x); evalf(2/3);
                                0., 0.6774193548
                                0.6666666667

```

(20)

```

> x:=0.7
                                x := 0.7

```

(21)

```

> for i from 1 to 1200 do x:=lambda*x*(1-x); f[i]:=x: od:
> points:=[[n,f[n]]$n=1..1200]:
> pointplot(points,symbol=circle);
pointplot([[1, 0.651], [2, 0.7043169], [3, 0.6455892736], [4, 0.7092916665], (22)
[5, 0.6392106948], [6, 0.7149231856], [7, 0.6318048752], [8,
0.7211451721], [9, 0.6233939200], [10, 0.7277992156], [11,
0.6141333038], [12, 0.7346181259], [13, 0.6043584384], [14,
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```

```
> restart;
```

```
> # Problem 6
```

```
> x:=0.67;
```

$$x := 0.67 \quad (23)$$

```
> y:=0.67001;
```

$$y := 0.67001 \quad (24)$$

```
> # chaotic dynamic => the difference between the ?solutions? is
very big even though the x and y are very close to each other
```

```
> for i
```

```
> for i from 1 to 40 do x:=4*x*(1-x): y:=4*y*(1-y): print(i,x,y);
od:
```

```
Error, reserved word `for` unexpected
```

```
> # b)
```

```
> for i frm 1 to 40 do x:=2*x*(1-x): y:=2*y*(1-y): z:=4*z-4*z*z:
print(i,x,y,z): od:
```

```
Error, missing operator or `:`
```

```
> #improved euler
```

```
> restart;
```

```
> # Pb 7
```

```
> restart;
```

```
> f:=(t,x)->2*t*x; dsolve({diff(x(t),t)=f(t,x(t)),x(0)=1}); phi:=
unapply(rhs(%),t);
```

$$f := (t, x) \mapsto 2 \cdot t \cdot x$$

$$x(t) = e^{t^2}$$

$$\phi := t \mapsto e^{t^2}$$

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$$\begin{aligned} & \text{> } h:=0.1; \text{ } x:=1; \text{ } t:=0; \\ & \quad h := 0.1 \\ & \quad x := 1 \\ & \quad t := 0 \end{aligned} \tag{26}$$

```
> #euler's method:
> for i from 1 to 10 do s:=x+h*f(t,x): t:=t+h: print(t,evalf(x),
  evalf(phi(t)),evalf(abs(x-phi(t)))); od:
0.1, 1., 1.010050167, 0.010050167
0.2, 1., 1.040810774, 0.040810774
0.3, 1., 1.094174284, 0.094174284
0.4, 1., 1.173510871, 0.173510871
0.5, 1., 1.284025417, 0.284025417
0.6, 1., 1.433329415, 0.433329415
0.7, 1., 1.632316220, 0.632316220
0.8, 1., 1.896480879, 0.896480879
0.9, 1., 2.247907987, 1.247907987
1.0, 1., 2.718281828, 1.718281828
```

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```
> # the output above might be wrong
> # improved euler:
> for i from 1 to 10 do x:=x+h/2*f(t,x)+h/2*f(t+h,x+h*f(t,x)): t:=
t+h: print(t,x,phi(t),abs(x-phi(t))); od:
1.1, 1.232000000, 3.353484653, 2.121484653
1.2, 1.547884800, 4.220695817, 2.672811017
1.3, 1.983150006, 5.419480705, 3.436330699
1.4, 2.590787168, 7.099327065, 4.508539897
1.5, 3.450928508, 9.487735836, 6.036807328
1.6, 4.686360914, 12.93581732, 8.249456406
1.7, 6.487798049, 17.99330960, 11.50551155
1.8, 9.155580606, 25.53372175, 16.37814114
1.9, 13.16938715, 36.96605281, 23.79666566
2.0, 19.30632156, 54.59815003, 35.29182847
```

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```

=> # last one should be 0.00922...
=> # hw pb 8
=> # Problem 9
=> restart;
> f:=(t,x)->-250*x;

```

$f := (t, x) \mapsto -250 \cdot x$
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$$\text{> dsolve(\{diff(x(t),t)=f(t,x(t)),x(0)=1\},x(t));}$$

$$x(t) = e^{-250t} \quad (30)$$

```
> phi:=unapply(rhs(%),t);
```

$$\phi := t \mapsto e^{-250 \cdot t}$$

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```
> h:=0.1; x:=1; t:=0;
```

$$h := 0.1$$

$$x := 1$$

$$t := 0$$

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```
> # Improved euler's:
> for i from 1 to 10 do x:=x+h/2*f(t,x)+h/2*f(t+h,x+h*f(t,x)): t:=
t+h: print(t,x,phi(t),abs(x-phi(t))); od:
0.1, 288.5000000, 1.388794386 × 10-11, 288.5000000
0.2, 83232.25000, 1.928749848 × 10-22, 83232.25000
0.3, 2.401250412 × 107, 2.678636962 × 10-33, 2.401250412 × 107
0.4, 6.927607438 × 109, 3.720075976 × 10-44, 6.927607438 × 109
0.5, 1.998614746 × 1012, 5.166420633 × 10-55, 1.998614746 × 1012
0.6, 5.766003544 × 1014, 7.175095973 × 10-66, 5.766003544 × 1014
0.7, 1.663492023 × 1017, 9.964733010 × 10-77, 1.663492023 × 1017
0.8, 4.799174487 × 1019, 1.383896527 × 10-87, 4.799174487 × 1019
0.9, 1.384561839 × 1022, 1.921947728 × 10-98, 1.384561839 × 1022
1.0, 3.994460906 × 1024, 2.669190216 × 10-109, 3.994460906 × 1024
```

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```
> # euler method for 9:
> #hw: 7 and 8
```